EDUHUB / ENGINEERING DOCUMENTATION

SEQUENCE DIAGRAMS

EduHub | Online Learning Platform

14	03	48
UML DIAGRAMS	APPLICATION LAYERS	USE CASE BASELINE

Message ordering, lifeline responsibilities, conditional branches, persistence boundaries, and external integrations.

Based on the supplied EduHub project brief (5 pages), with continuity to the previously prepared architecture, use-case, DFD, requirements, and schema documents.

SYSTEM ANALYSIS & DESIGN / NOVEMBER 2026

Interaction model | executive view

14 focused UML sequence views across EduHub user journeys and three application layers.

14	03	08	01
SEQUENCE VIEWS	LOGICAL LAYERS	DATA DOMAINS	SOURCE BRIEF

DELIVERABLE BOUNDARY

This document shows chronological interactions among actors, MVC controllers, business services, repositories, databases and external dependencies. It complements - not replaces - the Use Case Descriptions, ERD, Data Flow Diagram, logical/physical schema and API contracts.

WHAT IS MODELED

Successful and exceptional interactions; synchrony versus event-style signals; identity and ownership checks; EF Core persistence points; private media access and Stripe acknowledgment.

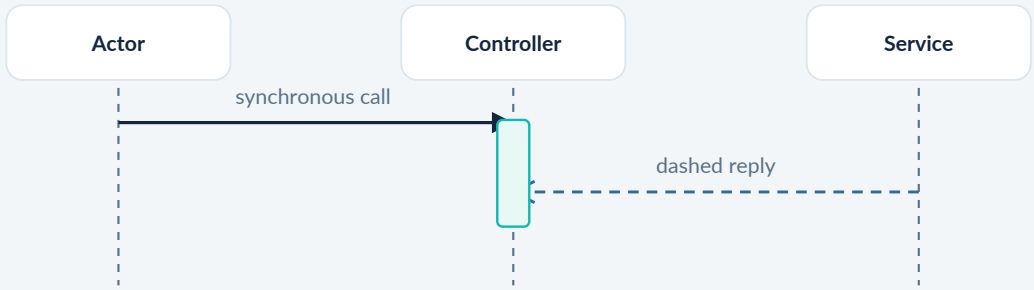
NOT CLAIMED AS IMPLEMENTED

Specific controller names, method calls, worker placement, selected database and asynchronous processing are illustrative design proposals, not evidence that the system already exists.

* Domain grouping follows the preceding ERD. Interaction views address more than one database domain.

Sequence notation & reading guide

UML 2.5.1 interaction concepts; scoped to reviewable implementation-facing sequences. [R1]



INTERACTION SEMANTICS

Solid arrow: request. Dashed arrow: response. Open asynchronous arrow: event signal. Small execution bar: illustrative receiver activation.

COMBINED FRAGMENT NOTATION

FRAGMENT	MEANING	BUSINESS EXAMPLE
alt	Exactly one guard is chosen; else branch is alternative.	Valid credentials versus sign-in denial.
opt	Optional subinteraction; used only when guard holds.	Notify learner when a certificate is issued.
loop	Repeated operation, with guard or bounds.	A cart contains multiple priced course items.

Notation note: events are ordered top-to-bottom; a numbered message is local to that diagram.

Event handlers are internal components - not additional human actors. Diagram lifelines are logical services.

Lifelines & N-Tier responsibilities

All diagrams retain the agreed modular monolith and three logical project boundaries. [S1, P1, R2]

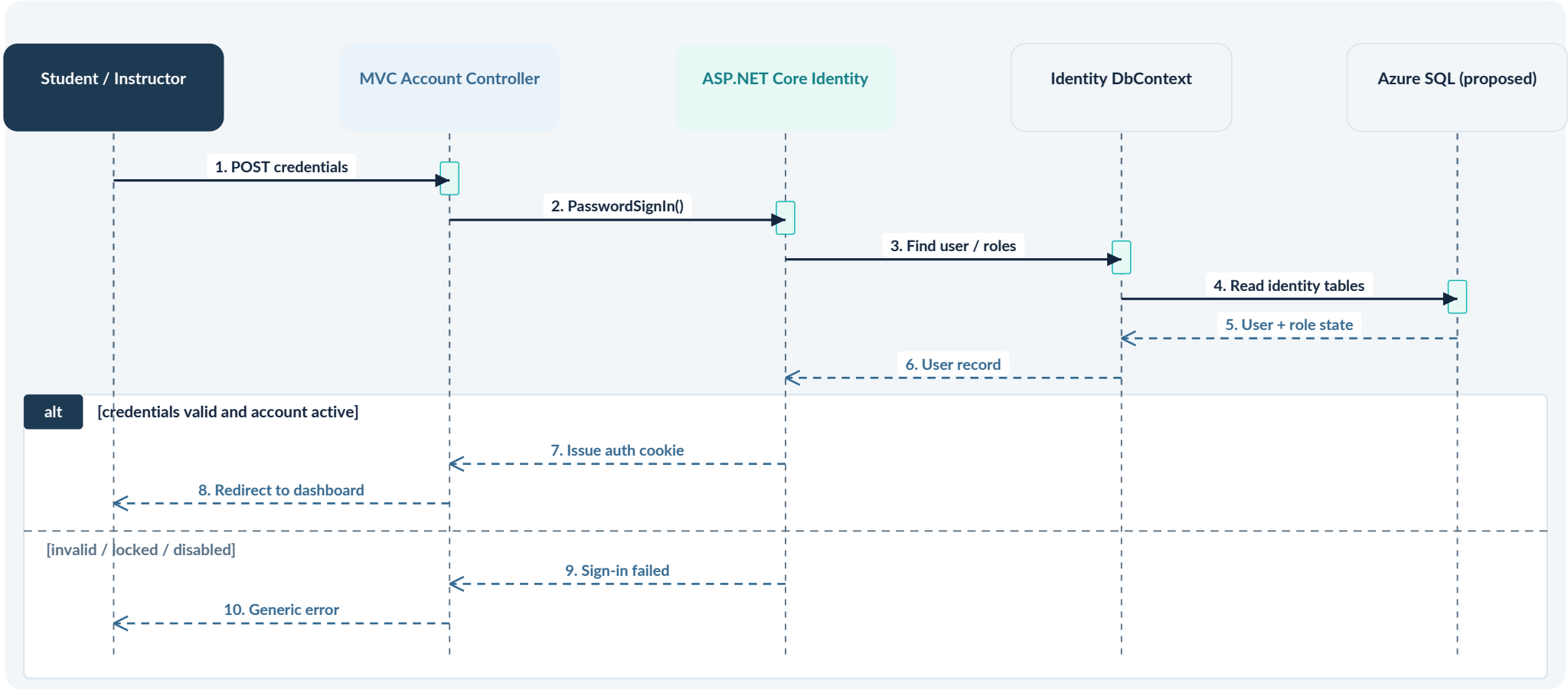
LAYER / PARTICIPANT	REPRESENTATIVE LIFELINES	ARCHITECTURAL RESPONSIBILITY
Presentation | EduHub.Web	Browser, Razor/Video.js, MVC controllers, Stripe webhook endpoint	Transport handling, session context, form validation, view models and responses.
Business | EduHub.Services	Course, Payment, Learning, Quiz, Certificate, Reporting services	Business rules, ownership/policy guards, orchestration, idempotency and integration ports.
Data | EduHub.Data	Repositories, Unit of Work, Identity DbContext, Azure SQL	EF Core mapping, queries, atomic commits, unique constraints and concurrency control.
External systems	Stripe Checkout, Azure Blob Storage, Azure CDN, LinkedIn share UI	Payment, private video/object storage and browser-initiated sharing.
Optional async adapter	Media processor / progress handler / notification handler	Proposed non-blocking follow-up work; may run in-process or via a separate worker.

INVARIANT

The Web layer should not query the database directly or decide payment eligibility; services enforce policy, repositories persist it, and external side effects remain outside the SQL transaction.

Authenticate a member

SD-01 | IDENTITY | UC-02, UC-06 | Brief + derived guards



NOMINAL OUTCOME

Authenticated session with server-side role policies; no privileged controller access without authorization.

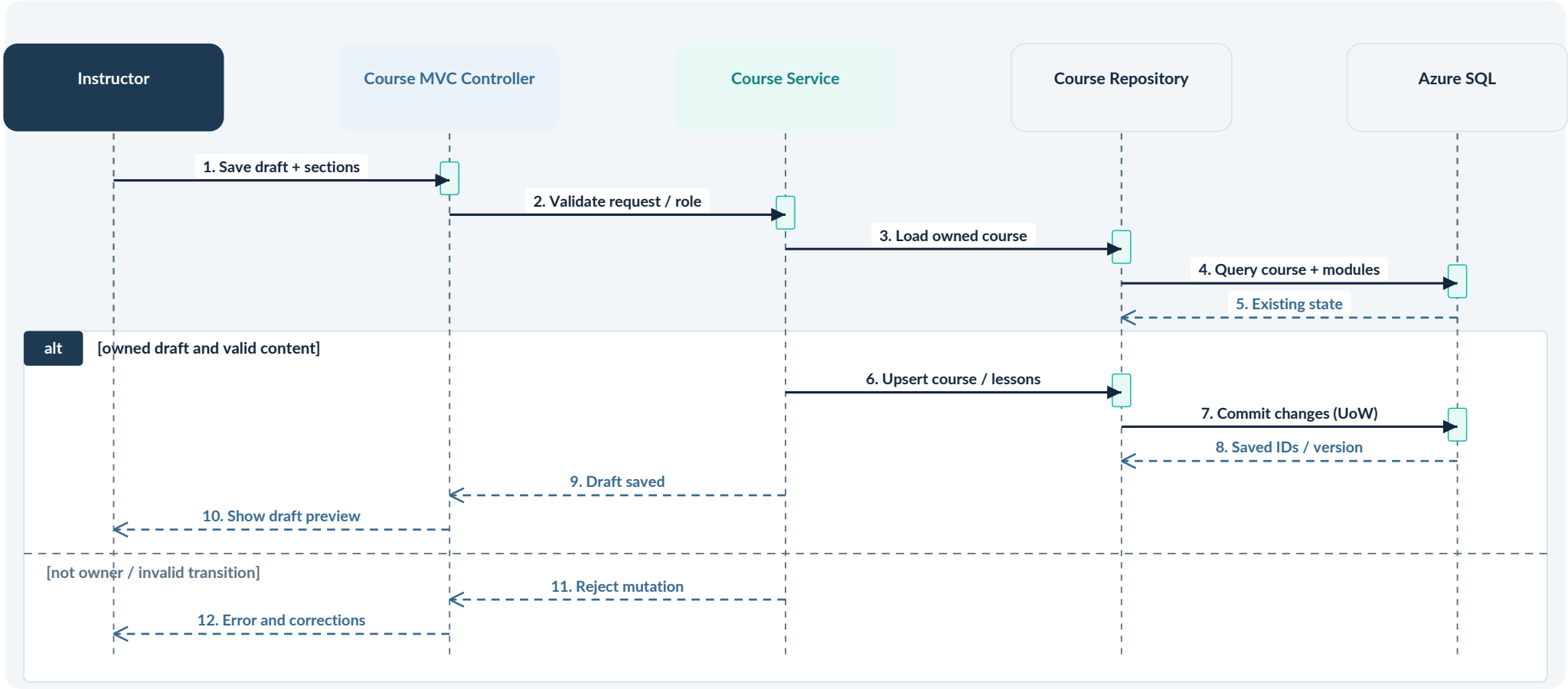
EXCEPTION / PROTECTION

Invalid or locked credentials never create a session. Keep error messages non-enumerating.

Trace: UC-02, UC-06 / FR-002, FR-003, FR-006

Save a course draft

SD-02 | COURSE AUTHORING | UC-25, UC-26, UC-29 | Brief + ownership safeguards



NOMINAL OUTCOME

Course, sections and lessons remain Draft until explicit submission and moderation.

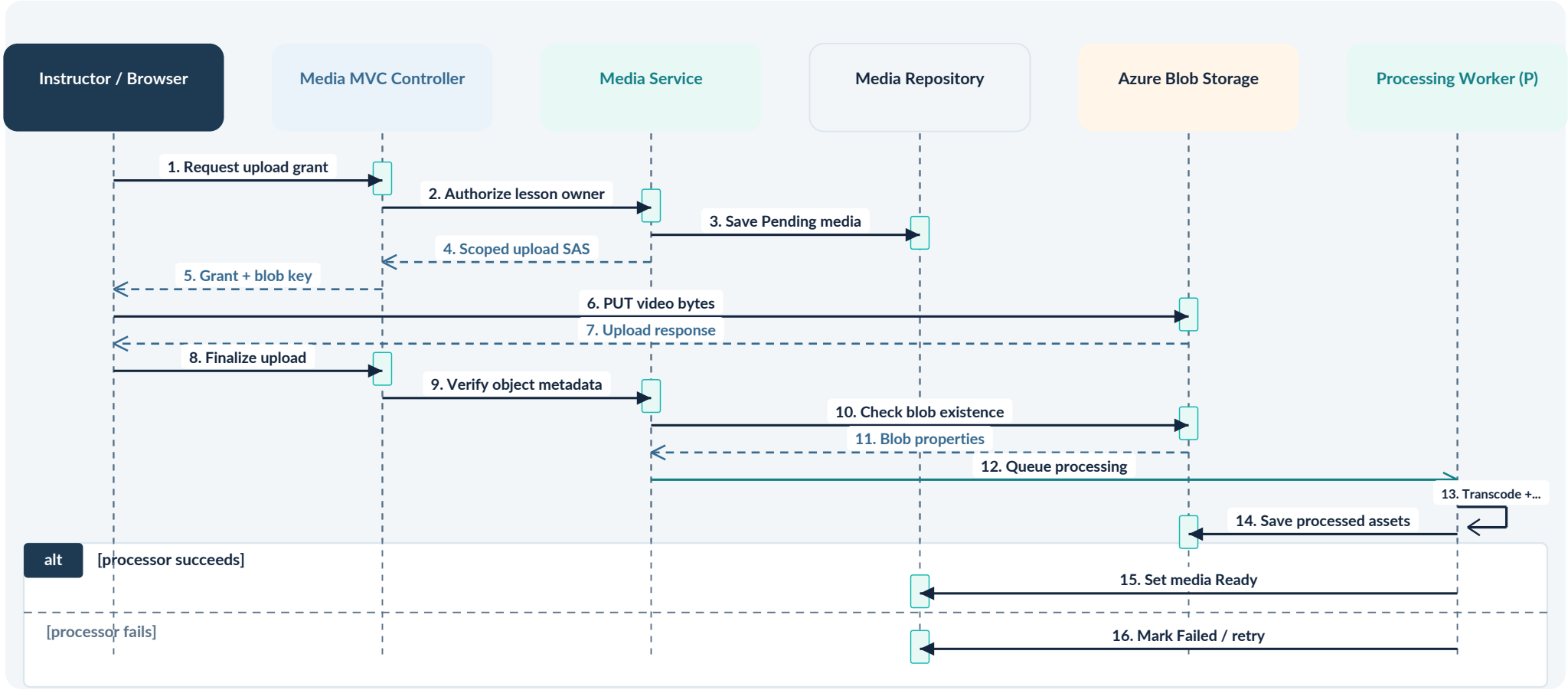
EXCEPTION / PROTECTION

The ownership check occurs in the service, not only in the Razor UI.

Trace: UC-25, UC-26, UC-29 / FR-014 to FR-019

Upload & process lesson video

SD-03 | MEDIA PIPELINE | UC-27, UC-46 | Brief + proposed worker



NOMINAL OUTCOME

Media metadata changes Pending -> Processing -> Ready / Failed; raw video bytes bypass SQL.

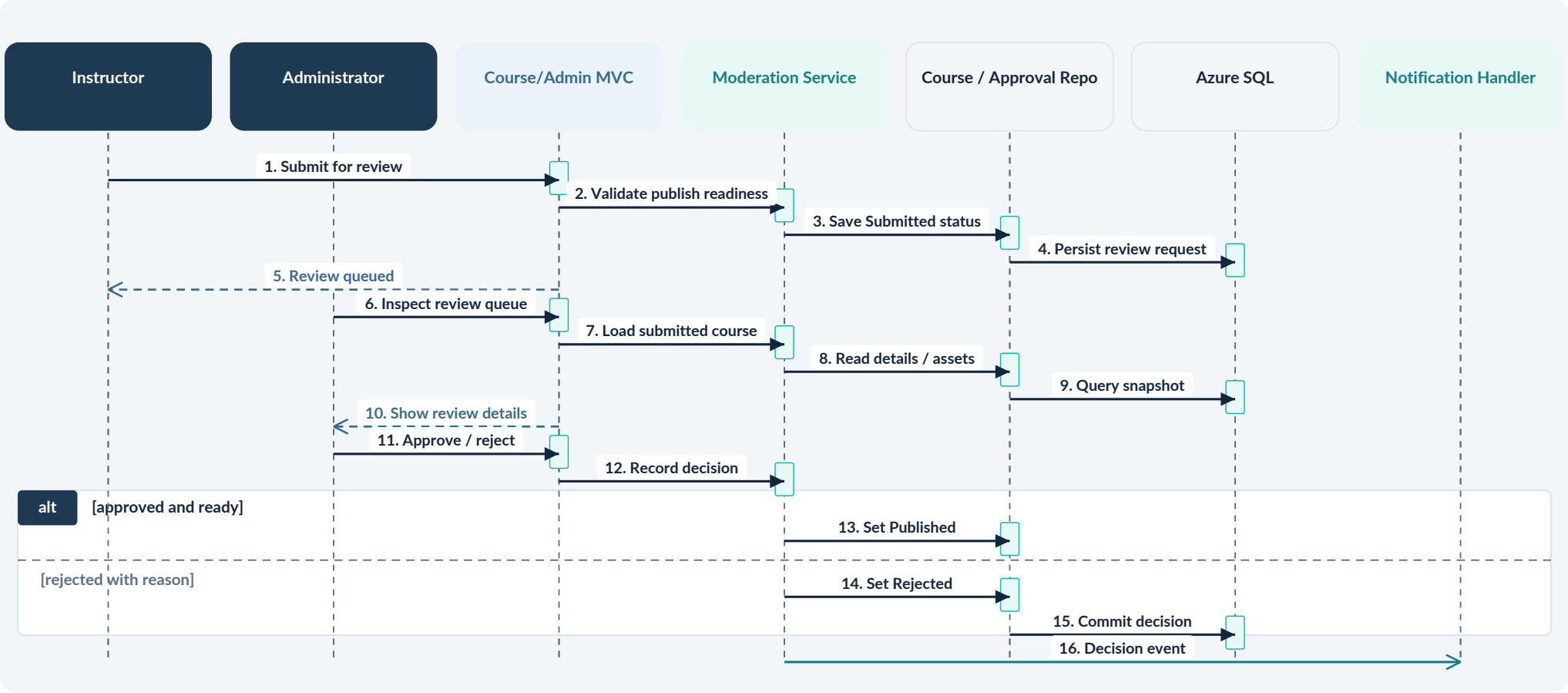
EXCEPTION / PROTECTION

Never trust a client-reported upload without verifying its blob and ownership.

Trace: UC-27, UC-46 / FR-020 to FR-023

Submit & approve a course

SD-04 | GOVERNANCE | UC-31, UC-37, UC-38, UC-48 | Moderation required; workflow proposed



NOMINAL OUTCOME

Only approved, valid courses appear publicly; moderation decision is durably recorded.

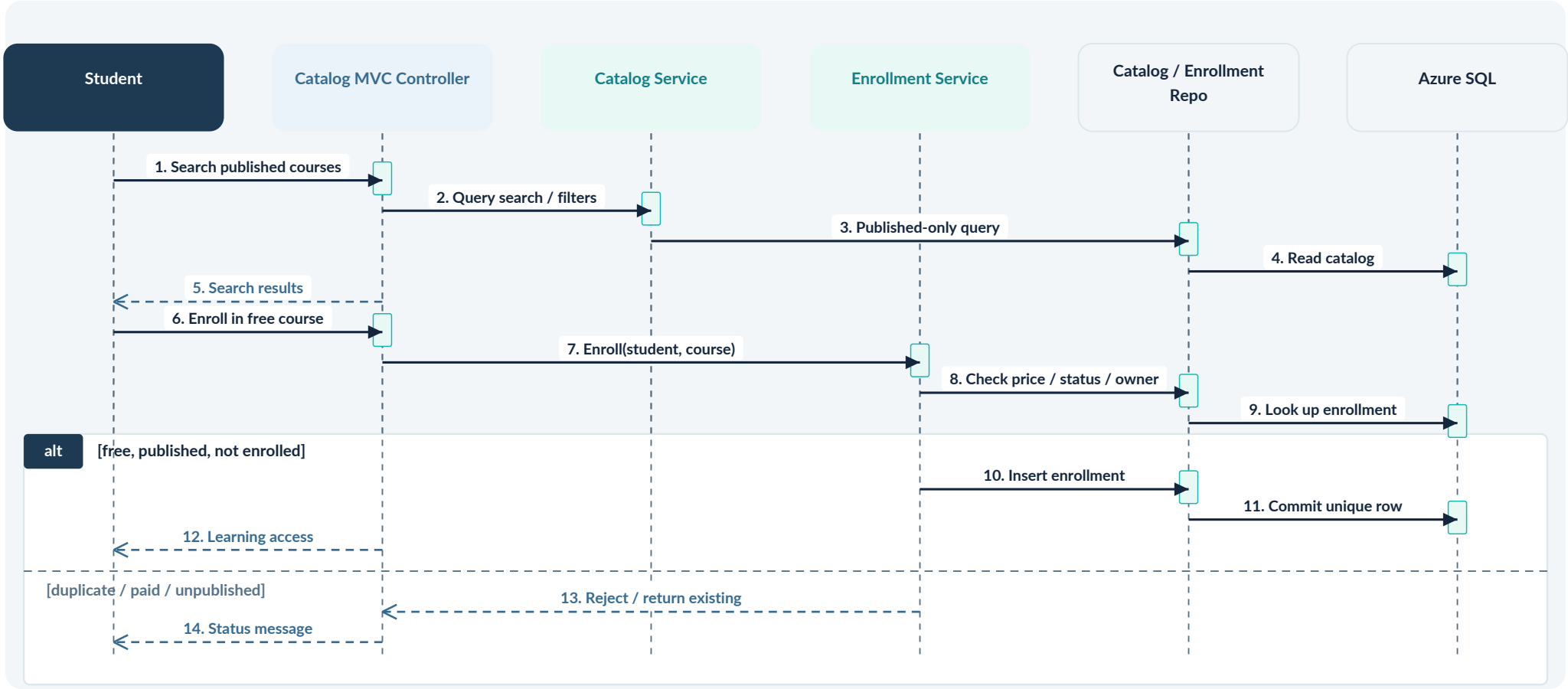
EXCEPTION / PROTECTION

Rejection must preserve a reason. Duplicate admin decisions require transition checks.

Trace: UC-31, UC-37, UC-38, UC-48 / FR-024, FR-069, FR-070

Browse & enroll in a free course

SD-05 | CATALOG / ENROLLMENT | UC-03, UC-04, UC-07 | Brief + uniqueness guarantee



NOMINAL OUTCOME

One enrollment per student-course pair; no Stripe interaction for free enrollment.

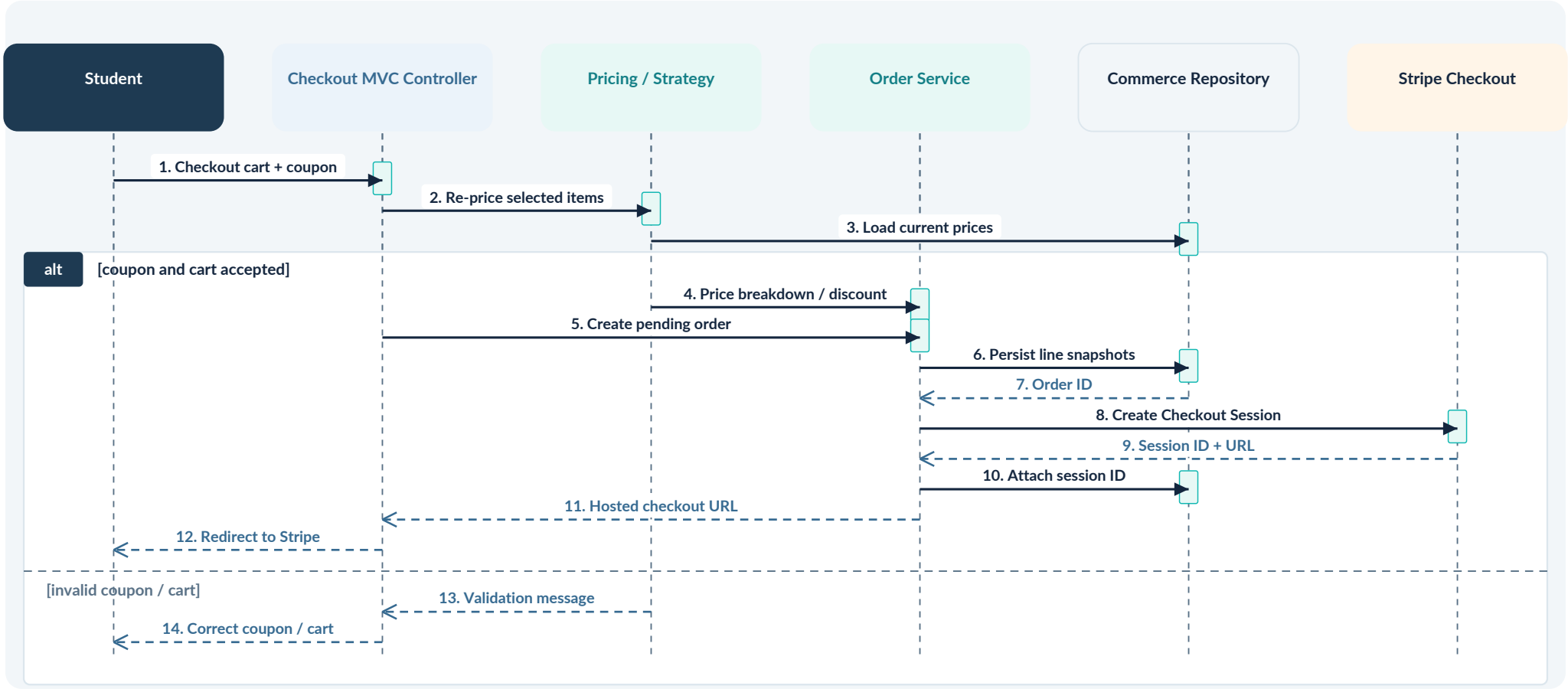
EXCEPTION / PROTECTION

The server rechecks course status and price before writing; UI price is not authoritative.

Trace: UC-03, UC-04, UC-07 / FR-007 to FR-012, FR-049

Build a paid Stripe checkout

SD-06 | COMMERCE | UC-08, UC-09, UC-10 | Brief + proposed price snapshot



NOMINAL OUTCOME

Pending order and immutable price snapshot are stored; no paid enrollment is granted yet.

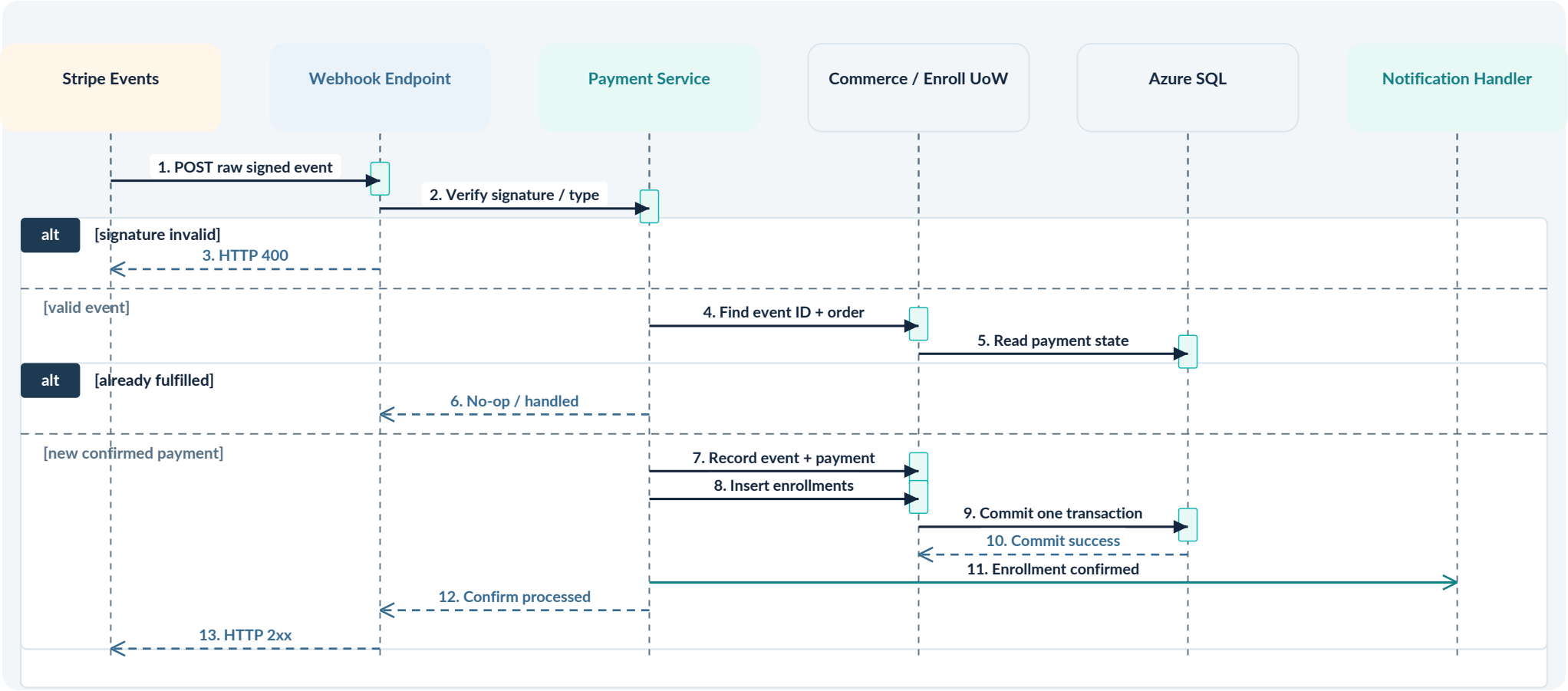
EXCEPTION / PROTECTION

A canceled checkout stays unfulfilled. Client redirects and UI amounts are never proof of payment.

Trace: UC-08, UC-09, UC-10 / FR-045 to FR-053

Verify webhook & fulfill an order

SD-07 | PAYMENT INTEGRITY | UC-43, UC-44, UC-48 | Critical proposed implementation



NOMINAL OUTCOME

Verified paid orders create entitlements exactly once, even after Stripe retries or duplicate delivery.

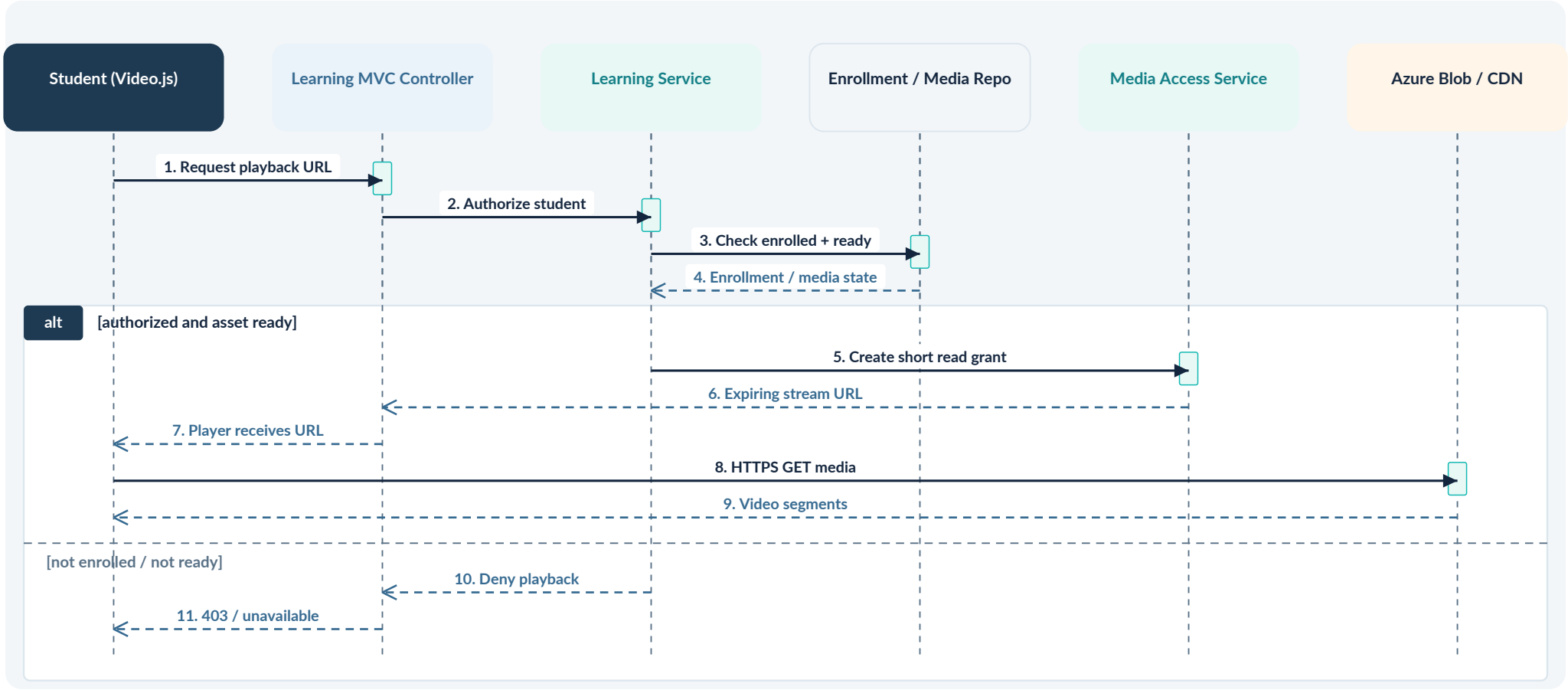
EXCEPTION / PROTECTION

On transient database failure, do not acknowledge uncommitted work as processed; permit safe retries.

Trace: UC-43, UC-44, UC-48 / FR-053 to FR-055

Authorize protected lesson playback

SD-08 | SECURE LEARNING | UC-12, UC-13, UC-47 | Brief + scoped URL proposal



NOMINAL OUTCOME

Binary media streams from blob/CDN; the application never exposes a permanent public video URL.

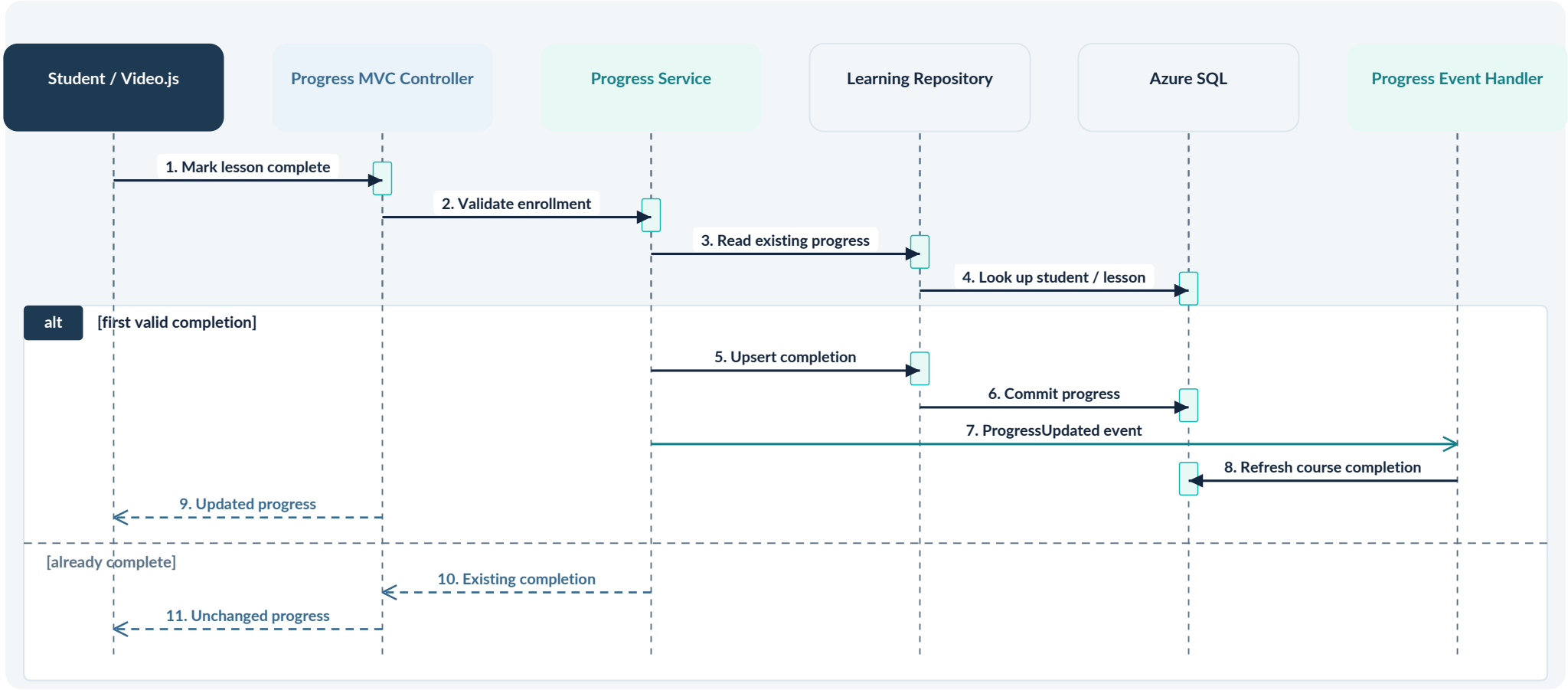
EXCEPTION / PROTECTION

Reissue a new short-lived grant after expiry; do not embed cloud storage credentials in the page.

Trace: UC-12, UC-13, UC-47 / FR-026 to FR-031

Record lesson completion

SD-09 | PROGRESS TRACKING | UC-14, UC-15, UC-48 | Brief + proposed event boundary



NOMINAL OUTCOME

Progress percentage derives from committed lesson records and published completion rules.

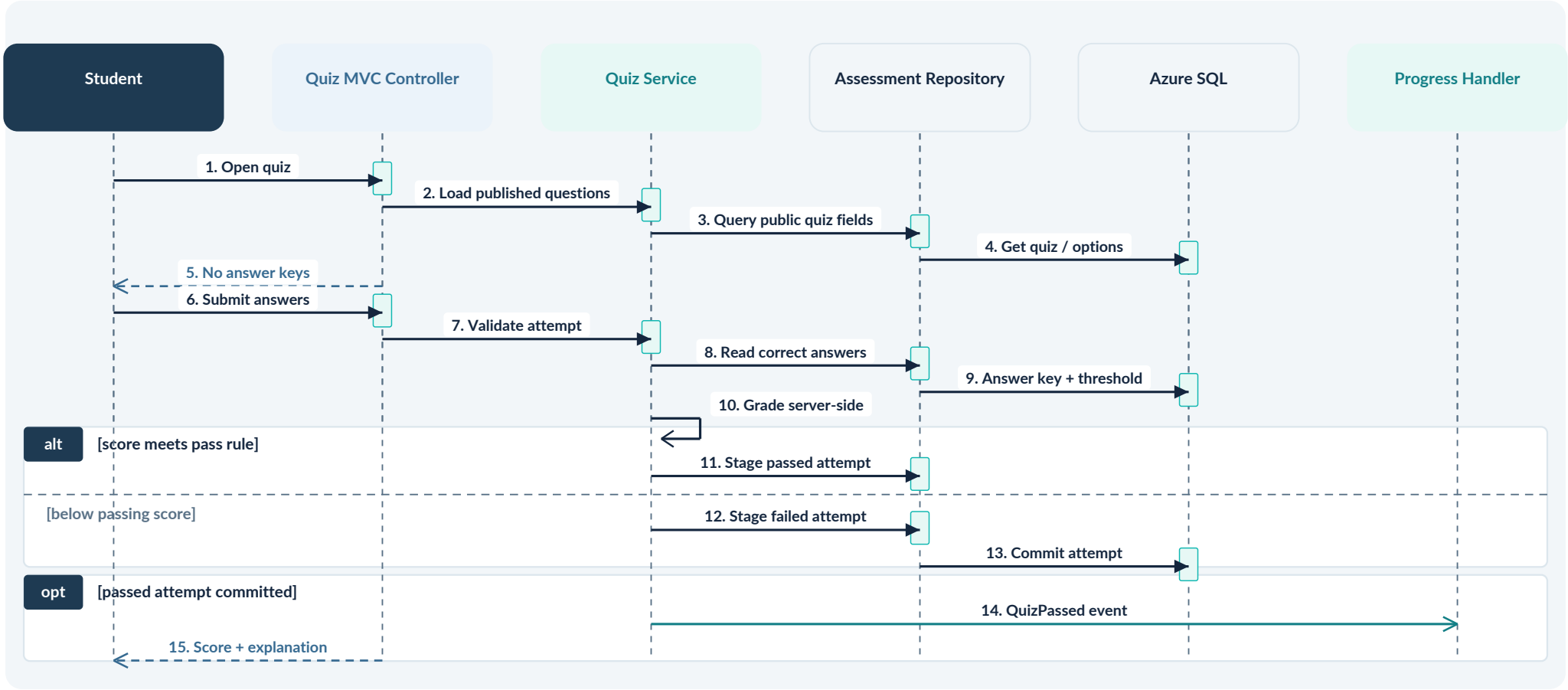
EXCEPTION / PROTECTION

Repeated or simultaneous requests cannot count one lesson twice.

Trace: UC-14, UC-15, UC-48 / FR-032 to FR-036

Attempt & grade a quiz

SD-10 | ASSESSMENT | UC-16, UC-33 | Brief + proposed retake policy



NOMINAL OUTCOME

An authoritative attempt record is stored with score, pass flag and permitted feedback.

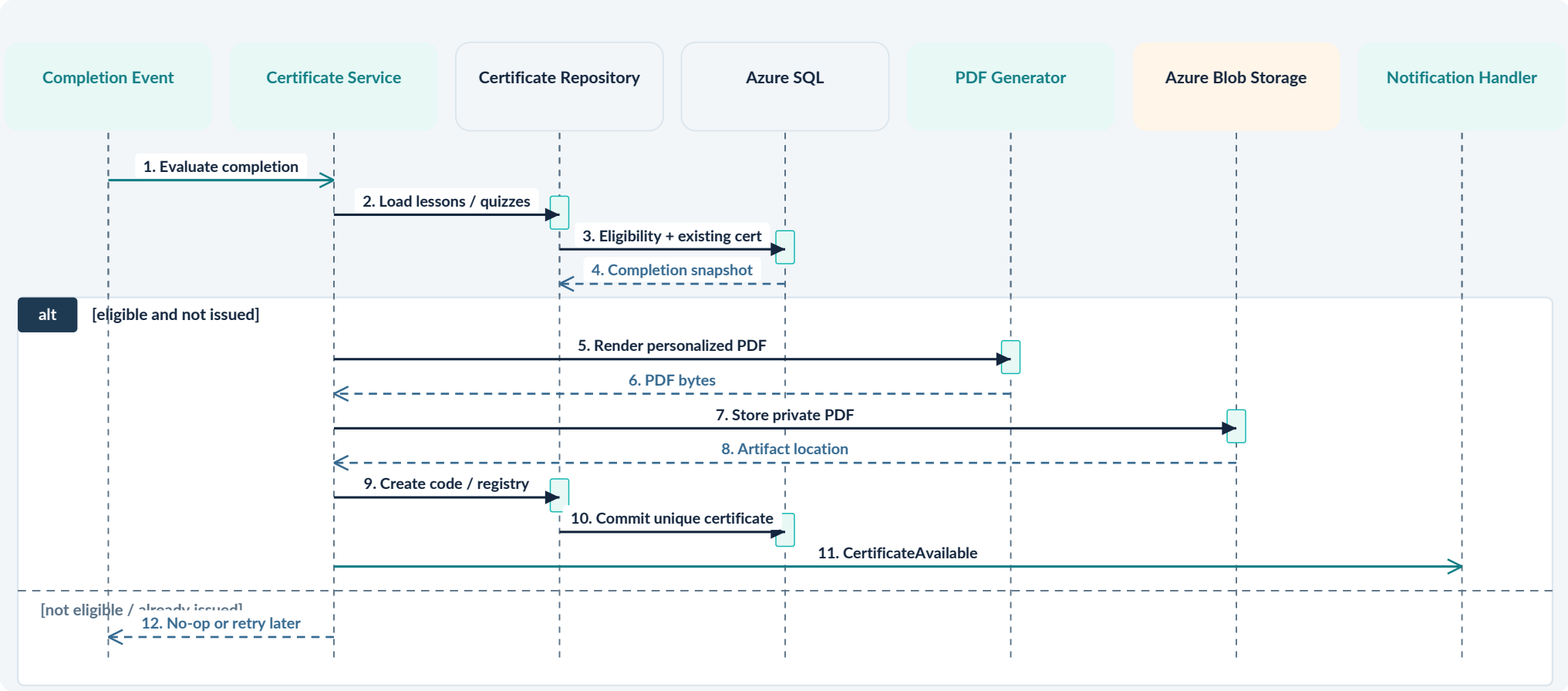
EXCEPTION / PROTECTION

The browser must not receive correct answers before submission; retries depend on approved policy.

Trace: UC-16, UC-33 / FR-037 to FR-044

Issue a completion certificate

SD-11 | CERTIFICATION | UC-23, UC-45 | Brief + proposed worker



NOMINAL OUTCOME

A unique certificate verification code is associated with one earned student-course result.

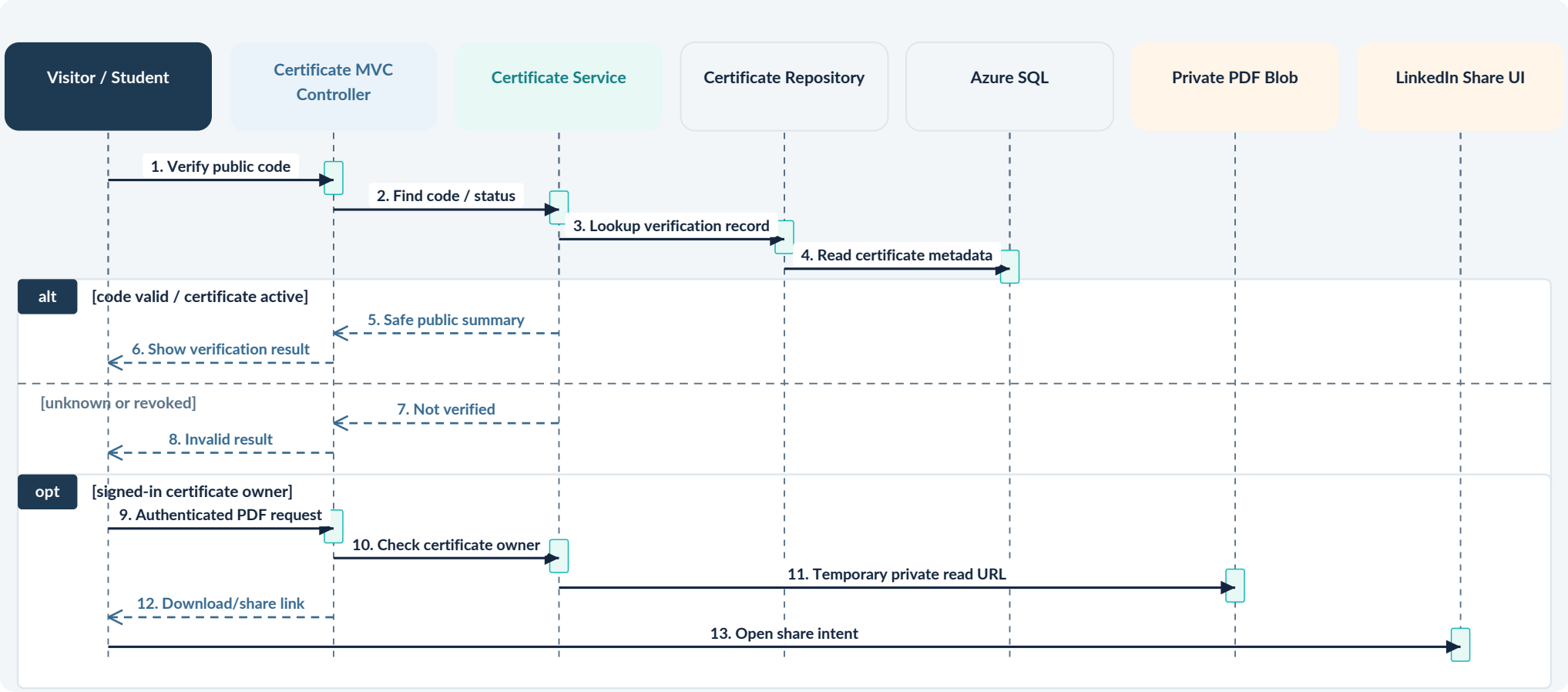
EXCEPTION / PROTECTION

Unique database constraints and retry behavior prevent duplicate certificates.

Trace: UC-23, UC-45 / FR-063 to FR-066

Verify & share a certificate

SD-12 | PUBLIC CERTIFICATES | UC-05, UC-23, UC-24 | Brief + minimal public lookup



NOMINAL OUTCOME

Public users see a minimal validity result; owners can retrieve a private PDF and share its verification link.

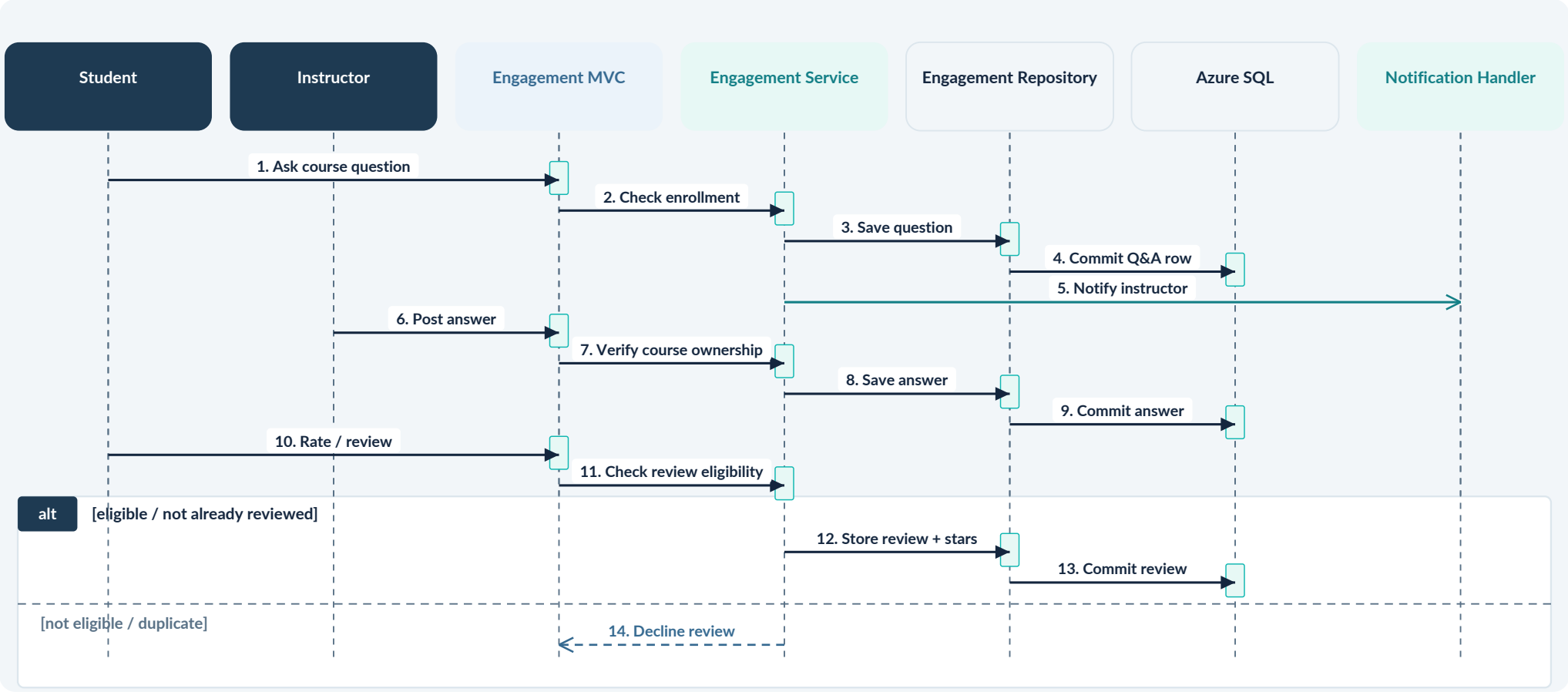
EXCEPTION / PROTECTION

The public endpoint must not expose user emails, identity records or permanent download URLs.

Trace: UC-05, UC-23, UC-24 / FR-065 to FR-068

Questions, answers & reviews

SD-13 | ENGAGEMENT | UC-19, UC-21, UC-22, UC-35 | Brief + proposed review eligibility



NOMINAL OUTCOME

Q&A and ratings remain owned and course-scoped; relevant notifications follow durable changes.

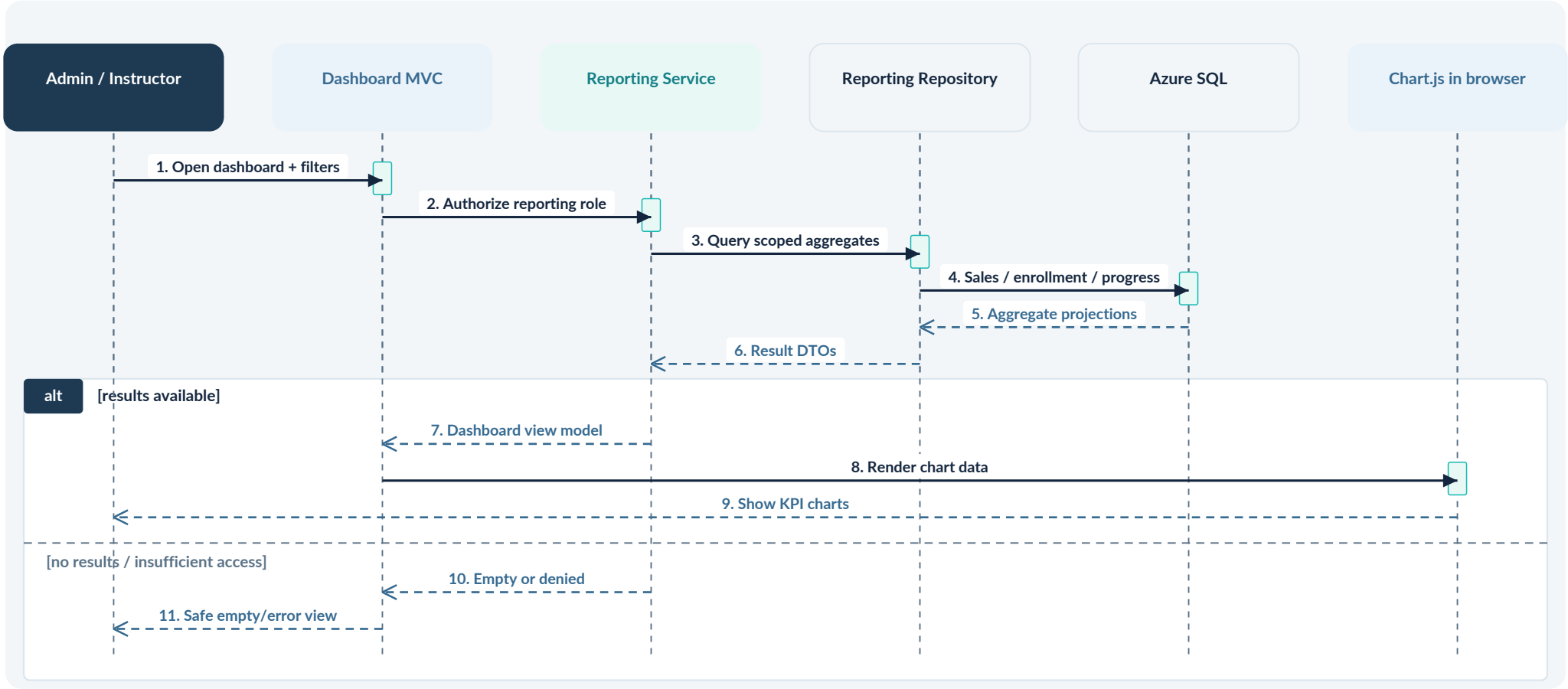
EXCEPTION / PROTECTION

Review helpful votes are separately idempotent, backed by one vote per user-review pair.

Trace: UC-19, UC-21, UC-22, UC-35 / FR-056, FR-057, FR-060 to FR-062

Display analytics dashboards

SD-14 | ANALYTICS | UC-34, UC-41 | Brief + role-scope safeguard



NOMINAL OUTCOME

Charts visualize authorized aggregates without exposing other instructors' detailed data.

EXCEPTION / PROTECTION

All aggregation scopes are enforced server-side; raw payment credentials are never included.

Trace: UC-34, UC-41 / FR-073 to FR-077

Critical sequence review | paid checkout

Payment state transition, idempotency, and Stripe delivery boundaries. [R3, R4]

SEQUENCE INVARIANTS

ORDER	RULE	REASON / VERIFICATION
1. Price	Server recalculates current prices and promotion policy; persist a line-item snapshot.	Protect against client-side amount tampering and changes during checkout.
2. Pending state	Save a pending order before issuing a Stripe Checkout Session.	Provides a durable local correlation key and ownership context.
3. Callback	Verify raw payload with Stripe webhook signing secret before processing.	Untrusted HTTP requests must not change paid entitlements.
4. Success	Use a confirmed-success Stripe payment signal; browser return alone is not authoritative.	The redirect may not arrive or may be opened before settlement.
5. One commit	Deduplicate provider event; mark payment/order and insert enrollments atomically.	No paid enrollment without matching committed paid order.
6. Retry	2xx for committed/duplicate events; non-2xx for transient uncommitted failures.	Safe event redelivery without duplicate records.

SCOPE NOTE

The diagrams show business-level calls, not a final Stripe API contract. Event types, delayed-payment handling and Checkout configuration must be finalized during integration.

Critical sequence review | secure media

Metadata and media bytes move through different trust boundaries. [R5]

UPLOAD AND PLAYBACK GUARANTEES

BOUNDARY	REQUIRED VALIDATION	OUTCOME
Instructor -> app	Authenticated instructor; owns draft lesson; safe size/type restrictions.	Media record Pending; scoped upload permission.
Browser -> Azure	A restricted, expiring write grant; storage receives binary content.	Video bypasses application SQL and normal MVC uploads.
Finalize -> service	Verify blob identity / properties against expected media record.	Unverified or mismatched uploads are not playable.
Worker -> storage	Optional processor creates optimized video / thumbnail; record Ready or Failed.	Azure Blob is storage; transcoding needs another process/tool.
Student -> app	Authenticated, enrolled, published course, ready media.	Server issues short-lived private read grant.
Player -> Azure/CDN	Retrieve only approved object through HTTPS, grant expires.	Binary playback does not require a controller byte proxy.

PROPOSAL REQUIRING APPROVAL

This model prefers short-lived user-delegation SAS when applicable. The storage account, caching behavior, playback format and processor must be validated against team resources.

Critical sequence review | earned completion

Progress and quiz events are inputs; the certificate is a derived business result.

COMPLETION GUARD AND EFFECTS

GUARD / EVENT	SERVICE ACTION	PERSISTED RESULT
Lesson marked complete	Upsert one student-lesson record; recompute lesson completion.	LessonProgress and course percentage.
Quiz attempted	Score against server-held answers and required passing threshold.	QuizAttempt / QuizAnswer; passed flag.
Course completion signal	Verify all required lessons and quizzes; check existing certificate.	Eligible / not eligible decision.
Eligible + absent certificate	Generate PDF; create unique verification code; store record after artifact save.	One issued Certificate and private artifact pointer.
Duplicate / retry	Use a uniqueness constraint keyed to course and enrollment/learner.	Return existing certificate; no duplicate issue.
Verification request	Retrieve code/status; return minimal publicly verifiable facts.	No exposure of payment or private profile data.

CONSISTENCY CHOICE

Delivery of progress events, certificate work and notifications is shown as logical event-style interaction. An outbox/queue is a recommended enhancement, not part of the supplied brief or a committed component.

Traceability | diagrams to use cases

The cross-reference uses IDs established in the earlier EduHub Use Case Model and SRS.

DIAGRAM	USER / SYSTEM INTERACTION	USE CASES	FUNCTIONAL REQUIREMENTS
SD-01	Authenticate a member	UC-02, UC-06	FR-002, FR-003, FR-006
SD-02	Save a course draft	UC-25, UC-26, UC-29	FR-014 to FR-019
SD-03	Upload & process lesson video	UC-27, UC-46	FR-020 to FR-023
SD-04	Submit & approve a course	UC-31, UC-37, UC-38, UC-48	FR-024, FR-069, FR-070
SD-05	Browse & enroll in a free course	UC-03, UC-04, UC-07	FR-007 to FR-012, FR-049
SD-06	Build a paid Stripe checkout	UC-08, UC-09, UC-10	FR-045 to FR-053
SD-07	Verify webhook & fulfill an order	UC-43, UC-44, UC-48	FR-053 to FR-055
SD-08	Authorize protected lesson playback	UC-12, UC-13, UC-47	FR-026 to FR-031
SD-09	Record lesson completion	UC-14, UC-15, UC-48	FR-032 to FR-036
SD-10	Attempt & grade a quiz	UC-16, UC-33	FR-037 to FR-044
SD-11	Issue a completion certificate	UC-23, UC-45	FR-063 to FR-066
SD-12	Verify & share a certificate	UC-05, UC-23, UC-24	FR-065 to FR-068
SD-13	Questions, answers & reviews	UC-19, UC-21, UC-22, UC-35	FR-056, FR-057, FR-060 to FR-062
SD-14	Display analytics dashboards	UC-34, UC-41	FR-073 to FR-077

Validation matrix | integration tests

Illustrative scenarios to test the message ordering and guarded alternatives.

TEST REF	TRIGGER / SIMULATED CASE	EXPECTED OBSERVATION
IT-01 / SD-01	Wrong password; locked user; unprivileged account.	No auth cookie or privileged controller call.
IT-02 / SD-02	Attempt update of another instructor's draft.	Forbidden; DB unchanged.
IT-03 / SD-03	Upload incomplete blob / oversized file / worker failure.	Not Ready; denied playback; recoverable state.
IT-04 / SD-04	Admin approves stale or unready submission.	Conflict or validation rejection; no public release.
IT-05 / SD-05	Concurrent duplicate free enrollments.	One unique enrollment only.
IT-06 / SD-06	Invalid coupon or canceled checkout.	No paid entitlement created.
IT-07 / SD-07	Forged event / duplicate delivery / DB retry.	Invalid rejected; no double fulfillment; retry recoverable.
IT-08 / SD-08	Nonenrolled learner opens a copied media link.	No fresh grant; existing grant short-lived.
IT-09 / SD-09	Repeat progress complete action 10 times.	One completion counted; stable progress.
IT-10 / SD-10	Submitted answer tampering / below-threshold quiz.	Server grade; correct pass status.
IT-11 / SD-11	Duplicate completion events.	One certificate code per earned result.
IT-12 / SD-12	Unknown certificate / unauthorized PDF download.	Safe public result / private download rejected.
IT-13 / SD-13	Non-owner answers Q&A / duplicate review.	Rejected server-side; integrity preserved.
IT-14 / SD-14	Instructor requests another's earnings.	Role-scoped report only.

Open decisions & review checklist

Resolve these assumptions before translating the diagrams into implementation-level contracts.

QUESTIONS FOR SUPERVISOR AND TEAM SIGN-OFF

REF	APPROVAL POINT
A-01	Published edits: do they return to moderation or go live immediately?
A-02	Which async mechanism will process videos, progress events and certificate jobs?
A-03	Which payment methods/currencies and delayed-payment events are in scope?
A-04	What is the refund / cancellation effect on enrollment and certificates?
A-05	What is the quiz retry / partial-credit / fill-in-blank scoring policy?
A-06	Can a course include optional lessons, and how is completion computed?
A-07	Where are generated certificates stored, and which PDF library/license is approved?
A-08	Which SAS lifetime and media format meet actual hosting constraints?

REVIEW RESULT

A diagram is ready for implementation only once actors, entry points, success/exception ordering, authorization decisions, transaction boundaries and ownership of external effects are unambiguous.

Sources & modeling references

Source-derived features are distinguished from recommended engineering details.

[S1] EduHub project brief — Project 3: EduHub - Online Learning Platform (5-page user-supplied PDF).

Local supplied project requirements; primary source for features, stack and roles.

[P1] Prior design baseline — Software Architecture; ERD; Logical & Physical Schema; Use Cases; Requirements; DFD.

Internal companion documents; prior use-case and FR identifiers reused in diagrams.

[R1] OMG UML 2.5.1 — https://www.omg.org/spec/UML/2.5.1/

Sequence lifelines, messages, execution specifications, combined fragments.

[R2] Microsoft ASP.NET Core MVC — https://learn.microsoft.com/en-us/aspnet/core/mvc/overview

Controller, view/model boundaries and MVC request responsibilities.

[R3] Stripe Webhooks — https://docs.stripe.com/webhooks

Signature validation, event delivery and event-based integration.

[R4] Stripe Checkout fulfillment — https://docs.stripe.com/checkout/fulfillment

Server-side fulfillment after confirmed checkout rather than redirect alone.

[R5] Azure Storage SAS — https://learn.microsoft.com/en-us/azure/storage/common/storage-sas-overview

Short-lived delegated blob access and recommended user delegation SAS.

[R6] EF Core transactions — https://learn.microsoft.com/en-us/ef/core/saving/transactions

Atomic persistence and transaction behavior.

BASELINE / STATUS

Candidate interaction design for academic review; implementation details marked as proposed require team confirmation.